SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 2 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I _________Meghanadh Ambati (192372281)___________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Meghanadh

Annexure A

APPLICATION- BASED QUESTIONS

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by step.

b. Explain how the repeating keyword influences each character of the ciphertext.

a. Step-by-step encryption of "HELLO" using the keyword "KEY"

The **Vigenère cipher** encrypts each letter of the plaintext by shifting it according to the corresponding letter of the repeating keyword.

Step 1: Write the plaintext and repeat the keyword

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E

Step 2: Convert letters to numerical values (A = 0, B = 1, ..., Z = 25)

Letter	H	E	L	L	O
Plaintext Value	7	4	11	11	14

Letter	K	E	Y	K	E
Keyword Value	10	4	24	10	4

Step 3: Apply the encryption formula
Ciphertext = (Plaintext + Keyword) mod 26

Plaintext	7	4	11	11	14
Keyword	10	4	24	10	4
Sum	17	8	35	21	18
Mod 26	17	8	9	21	18

Step 4: Convert the results back to letters

Number	17	8	9	21	18
Letter	R	I	J	V	S

Encrypted Message (Ciphertext): RIJVS

b. How the repeating keyword influences the ciphertext

- The keyword **"KEY"** is repeated until it matches the length of the plaintext:
    - HELLO
    - KEYKE
- Each keyword letter determines the shift applied to the corresponding plaintext letter.
- Different keyword letters produce different shifts:
    - **K = +10**
    - **E = +4**
    - **Y = +24**
- Because the keyword repeats, the shift pattern also repeats, making the ciphertext more secure than a simple Caesar cipher, where the same shift is applied to every character.
- Without the keyword, recovering the original message is significantly more difficult.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.

b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

a. How investigators can detect hidden data in the image

Investigators can use the following techniques:

1. **Statistical Analysis**
    - Analyze pixel values and color distributions for unusual patterns. ○ Hidden data often alters the Least Significant Bits (LSBs) of pixels, creating detectable statistical anomalies.
2. **Steganalysis Tools**
    - Use forensic software such as StegExpose, Stegdetect, or OpenStego to scan the image for hidden information.
    - These tools identify common steganographic techniques and estimate the likelihood of embedded data.
3. **Metadata Examination**
    - Inspect file metadata for unusual information, timestamps, or software signatures that indicate image modification.
4. **File Size and Format Comparison**
    - Compare the suspect image with the original (if available).

- ○ Unexpected increases in file size or structural inconsistencies may indicate embedded data.

5. **Visual and Frequency Analysis**
   - ○ Enhance brightness, contrast, or examine image frequency components.
   - ○ Some hidden information may become visible after image processing.

b. Two techniques to strengthen the security of hidden communications

1. Encrypt the Secret Data Before Embedding

- Encrypt the confidential information using a strong encryption algorithm (such as AES) before hiding it in the image.
- Even if investigators extract the hidden data, they cannot read it without the encryption key.

Benefit: Protects the confidentiality of the hidden message even if it is detected.

2. Use Adaptive or Randomized Steganography

- Instead of embedding data sequentially, hide it in randomly selected pixels determined by a secret key.
- Adaptive techniques place data in textured or noisy regions where modifications are less noticeable.

Benefit: Makes hidden data much harder to detect through statistical analysis or steganalysis.

RUBRICS FOR CALCULATION:

Criteria	Weigh ta ge	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understa ndi ng of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/tech no lo gies	Good understa ndi ng with proper integration of most concepts	Basic understan din g; limited or partial integration	Poor understan din g; unable to integrate concepts
Experim ent al Design	30	Well-planned, systematic implementati on with optimal solution	Correct implemen tat ion with minor issues	Basic implementa ti on with errors or inefficiencies	Incorrect or incomplete implement ati on

Use of Tools	20	Effective and advanced use of tools/technol ogi es with proper configuration	Appropria te use of tools with correct execution	Limited or inconsiste nt use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	interpretatio n Basic analysis with limited	insights Poor or incorrect analysis of results
Documen tat ion	10	Clear, well structured documentation with diagrams, code, and results	Organized document ati on with necessary details	Basic documenta tio n with missing details	Poor or incomplete documenta tio n